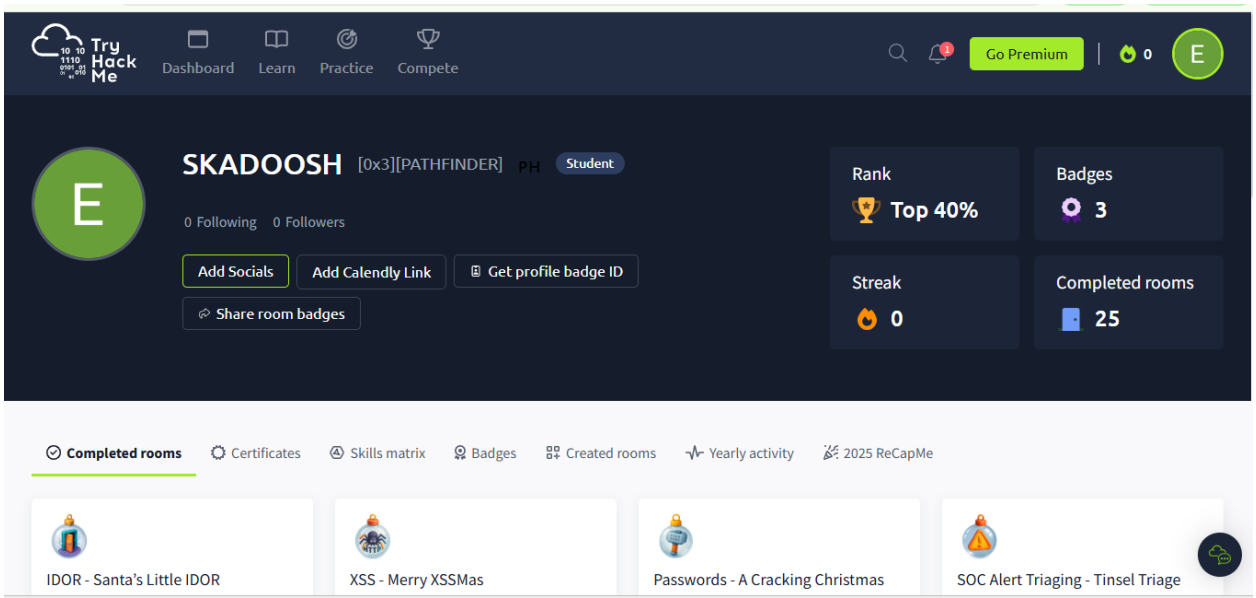
EDCEL REPOLLO
BSIT 3

LOVE AT FIRST BREACH – TryHackMe Event



Introduction

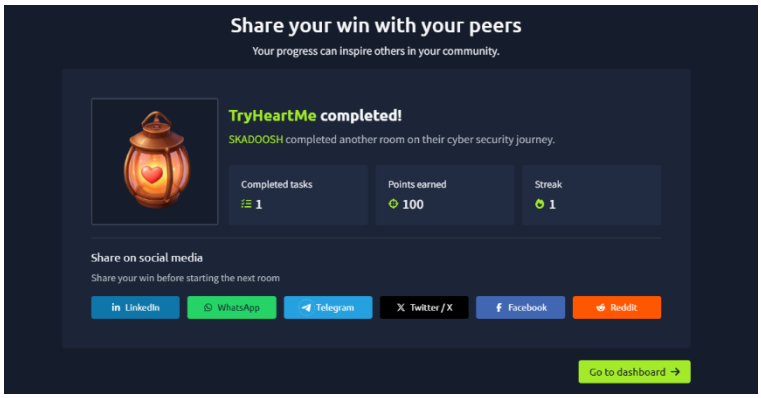
The Love at First Breach challenge on the TryHackMe platform is a Capture the Flag (CTF) activity designed to test participants' ability to analyze and exploit different security vulnerabilities in web applications. The event contains several hands-on challenges that simulate real-world cybersecurity scenarios such as authentication weaknesses, token manipulation, and improper access control.

Participating in this activity allowed me to apply the cybersecurity concepts I learned in class to practical situations. Each challenge required observation, problem-solving, and understanding how attackers can exploit vulnerabilities in poorly secured systems. Through this experience, I gained valuable insights into how web security flaws work and how they can be prevented.

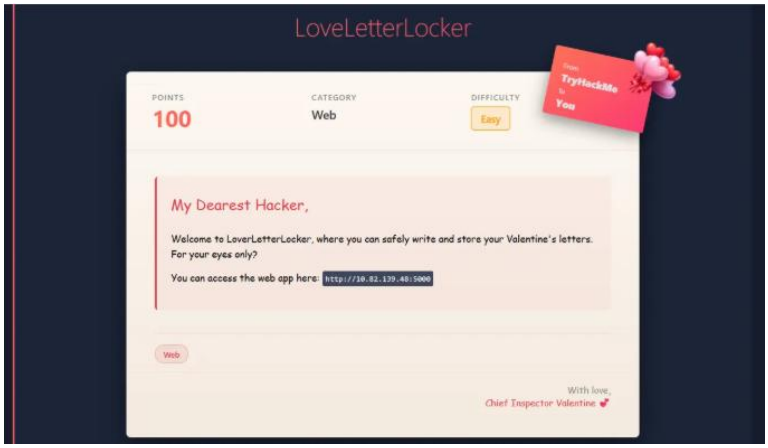
CHALLENGES COMPLETED

1. TryHeartME

Nothing says 'romance' like a successful exploit. Part of TryHackMe's Love at First Breach 2026 event, the TryHeartMe challenge drops us into a fictional e-commerce shop where the prices are high, but the security is low. In this walkthrough, we're going to play Cupid with the web traffic — intercepting requests and manipulating data to bypass the checkout and secure our flag. Whether you're here for the loot or the learning, let's dive into how we can break this shop's heart (and its code).

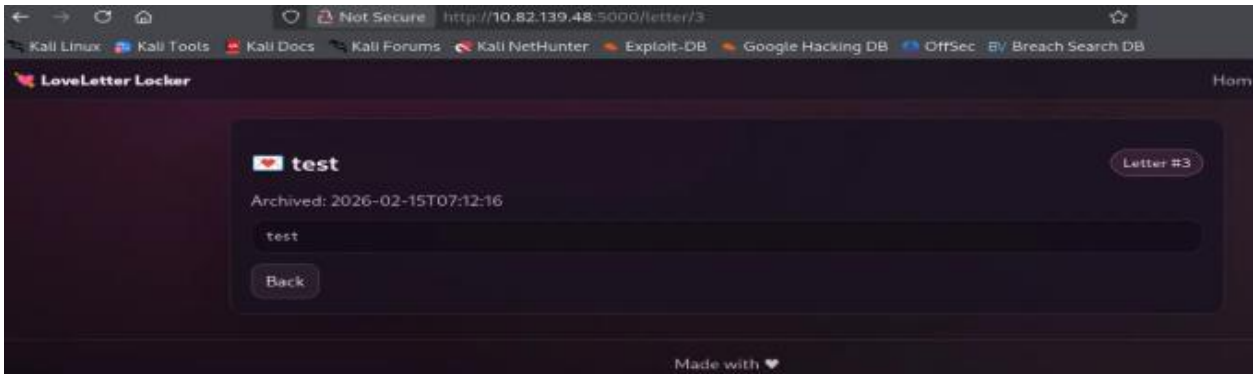


2. LOVE LETTER LOCKER



The **Love Letter Locker** challenge presents a web application where users can store personal letters securely. At first glance, the system appears to provide private storage for users' messages. However, further investigation revealed a vulnerability in how the application handled user data.

After registering an account and logging in, I noticed that the dashboard showed that several letters already existed in the system. When I created my own letter and opened it, I observed that the URL contained a numeric identifier such as:



To test whether the system properly verified access permissions, I manually changed the number in the URL from 3 to 1. Surprisingly, the system displayed a different letter that did not belong to my account. This confirmed that the application had an Insecure Direct Object Reference (IDOR) vulnerability.

Because the application did not verify whether the logged-in user owned the letter being requested, it allowed unauthorized access to other users' data.

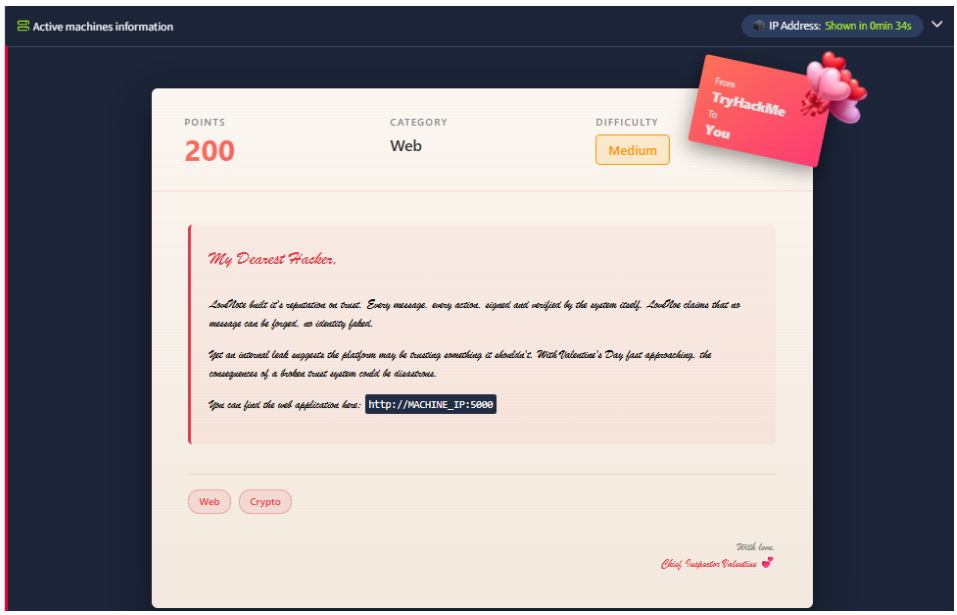
Inside the accessed letter, the hidden flag was revealed.

This challenge highlighted the importance of implementing proper authorization checks and access control mechanisms to ensure that users can only access their own data.

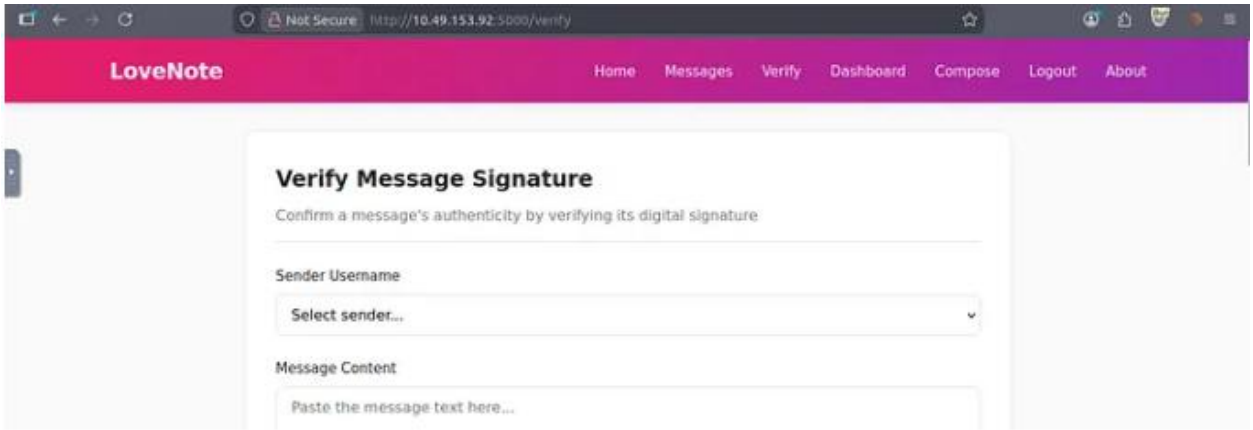
3. SIGNED MESSAGES

The Signed Messages challenge focused on understanding how digital signatures work in message authentication systems.

When accessing the application, the system allowed users to submit signed messages for verification. The goal was to identify weaknesses in how the application validated signed data.



After analyzing the structure of the signed message, I noticed that the signature was generated based on specific message parameters. By carefully modifying the message content and observing how the system responded, I was able to understand how the verification mechanism worked.



Through trial and analysis, it became clear that the system relied heavily on the integrity of the signed message structure. By manipulating parts of the message and testing different variations, I eventually managed to generate a valid signed message that revealed the hidden flag.

This challenge demonstrated the importance of **secure cryptographic implementation** and proper verification processes when handling digitally signed messages.

LEARNING INSIGHTS

Participating in these challenges helped me better understand several important cybersecurity concepts.

The **TryHeartMe** challenge taught me how authentication tokens such as JSON Web Tokens can be exploited when they are not securely validated by the server.

The **Love Letter Locker** challenge helped me understand how **Insecure Direct Object Reference (IDOR)** vulnerabilities occur when applications fail to verify user permissions properly.

The **Signed Messages** challenge introduced me to the importance of **cryptographic validation and secure message authentication**.

Overall, these exercises improved my analytical thinking and strengthened my ability to identify security weaknesses in web applications.

PERSONAL REFLECTION

Completing the challenges in the Love at First Breach event was both challenging and rewarding. Each task required patience, critical thinking, and experimentation in order to understand how the vulnerabilities worked.

Although some parts were difficult at first, the process of analyzing the system and discovering the weaknesses helped me improve my cybersecurity skills. It also gave me a better understanding of how attackers approach real systems and how developers must implement proper security measures.

CONCLUSION

The Love at First Breach challenges provided valuable hands-on experience in identifying and exploiting common web vulnerabilities. Through the challenges such as **TryHeartMe, Love Letter Locker, and Signed Messages**, I gained practical knowledge about authentication weaknesses, insecure access control, and cryptographic security.

These activities demonstrated how small security flaws can be exploited by attackers if proper protections are not implemented. They also emphasized the importance of secure coding practices, strong authentication mechanisms, and proper system configuration.

Overall, the experience strengthened my understanding of cybersecurity concepts and helped me develop practical skills that are essential in protecting modern web applications.